

Current	Result	Size	Time	CPU	GPU	SM Frequency	Process	Attributes
gemm_wmma_kernel	575	8,81x(128,4,1)	64.90 us	105,032	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 GHz	[19803] gemm_double_buffer	
Summary	Details	Source	Context	Comments	Raw	Session		

GPU Speed of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	22.80	Elapsed Cycles [cycle]	64.50
Memory Throughput [%]	29.92	Elapsed Cycles [cycle]	105,032
L1/TEX Cache Throughput [%]	30.36	SM Active Cycles [cycle]	102,012.88
L2 Cache Throughput [%]	10.61	SM Frequency [GHz]	1.62
DRAM Throughput [%]	9.52	DRAM Frequency [GHz]	7.98

This kernel grid is too small to fill the available resources on this device, resulting in only 0.9 full waves across all SMs. Look at [Launch Statistics](#) for more details.

Key Performance Indicators

Roofline Analysis

The ratio of peak float (FP32) to double (FP64) performance on this device is 64.1. The workload achieved close to 0% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the [Profiling Guide](#) for more details on roofline analysis.

Tensor Core Operations Roofline

Operations	# Operations / Cycle	# Operations / s	Peak % Peak Operations / s	Peak Operations / s
Srcfp16 Dstfp32 Sparsity=off	0	0	0	12,288 19,865,877,712,031.56
Srcfp16 Dstfp32 Sparsity=on	0	0	0	24,576 39,731,755,424,063.12
Srcfp16 Dstfp16 Sparsity=off	0	0	0	24,576 39,731,755,424,063.12
Srcfp16 Dstfp16 Sparsity=on	0	0	0	49,152 79,463,510,848,126.23
Srcfp16 Dstfp32 Sparsity=off	268,435,456	2,558.56	4,136,394,477,317.55	20.82 12,288 19,865,877,712,031.56
Srcfp16 Dstfp32 Sparsity=on	0	0	0	24,576 39,731,755,424,063.12
Srcfp64	0	0	0	96.54 139,900,847,267.83
Srcfp64 Sparsity=off	0	0	0	49,152 79,463,510,848,126.23
Srcfp64 Sparsity=on	0	0	0	98,304 158,927,021,696,252.
Srcint1	0	0	0	393,216 635,708,086,785,009.
Srcint4 Sparsity=off	0	0	0	98,304 158,927,021,696,252.
Srcint4 Sparsity=on	0	0	0	196,608 317,854,048,392,504.
Srcint8 Sparsity=off	0	0	0	49,152 79,463,510,848,126.23
Srcint8 Sparsity=on	0	0	0	98,304 158,927,021,696,252.
Srcf32 Dstfp32 Sparsity=off	0	0	0	6,144 9,932,938,856,015.78
Srcf32 Dstfp32 Sparsity=on	0	0	0	12,288 19,865,877,712,031.56

Key Performance Indicators

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us] 1 # Pass Groups 2

Maximum Buffer Size [Mbytes] 4

Overview

Average Active Warps Per Cycle	42.7 warp
Total Active Warps Per Cycle	1,02k warp
Blocks Launched	64 block
SM Active Cycles	1.63k cycle
Executed ipc Active	2.18 inst/cycle
SM ALU Pipe Throughput	100 %
SM FMA Light Pipe Throughput	100 %
SM FMA Heavy Pipe Throughput	100 %
SM Tensor Pipe Throughput	100 %
SM XU Pipe Throughput	100 %
DRAM	100 %
DRAM Throughput	0
DRAM Read Bandwidth	100 %
DRAM Write Bandwidth	100 %
L2 Cache	100 %
L2 Throughput	0
L2 Hit Rate	100 %
L1 Cache	100 %
L1 Throughput	466 cycle
L1 Writeback Throughput	0
L1 Hit Rate	100 %
L1 Wavefronts (Data)	1.22k
Workload Execution	gemm_wmma_kernel

Key Performance Indicators

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed ipc Elapsed [inst/cycle] 1.71 SM Busy [%] 42.80

Executed ipc Active [inst/cycle] 1.76 Issue Slots Busy [%] 42.80

Executed ipc Active [inst/cycle] 1.76

Balanced

ALU is the highest-utilized pipeline (33.8%) based on elapsed cycles in the workload, taking into account the rates of its different instructions. It executes integer and logic operations. It is well-utilized, but should not be a bottleneck.

Pipe Utilization (% of elapsed cycles)

Pipe Utilization (% of peak instructions executed over elapsed cycles)

Unit	Utilization [%]
ALU	33.8
FMA	25.0
Tensor (All)	10.0
Tensor (FP)	10.0
FP64	10.0
Tensor (INT)	10.0

Key Performance Indicators

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with the data for each memory unit.

Memory Throughput [byte/s] 24.32 Mem Busy [%] 28.38

L1/TEX Hit Rate [%]	34.01	Max Bandwidth [%]	29.52
L2 Hit Rate [%]	78.65	Mem Pipes Busy [%]	29.52
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	6.29
L1TEX Global Load Access Pattern	The memory access pattern for global loads from L1TEX might not be optimal. On average, only 30.2 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the [Source Counters](#) section for uncoalesced global loads.		
Est. Speedup: 1.64%			
	Key Performance Indicators		
L1TEX Global Store Access Pattern	The memory access pattern for global stores to L1TEX might not be optimal. On average, only 16.0 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the [Source Counters](#) section for uncoalesced global stores.		
Est. Speedup: 14.76%			
	Key Performance Indicators		

Memory Chart

Values: Transfer Size Inactivity: Greyed Out

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Selected Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp] 10.34 No Eligible [%] 56.31

Eligible Warps Per Scheduler [warp] 1.17 One or More Eligible [%] 43.69

Issued Warp Per Scheduler 0.44

Issue Slot Utilization

Est. Speedup: 56.31%

Est. Speedup: 56.31%

Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 2.3 cycles. This might have hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this workload allocates an average of 10.34 active warps per scheduler, but only an average of 1.17 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the [Warp State](#) and [Warp State](#) sections can help, too.

Key Performance Indicators

Warps Per Scheduler

GPU Maximum Warps Per Scheduler

Theoretical Warps Per Scheduler

Active Warps Per Scheduler

Eligible Warps Per Scheduler

Issued Warp Per Scheduler

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the overall performance. The higher the value, the more warp parallelism is required to hide this latency. For each warp state the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle] 23.67 Avg. Active Threads Per Warp 32

Warp Cycles Per Issued Instruction [cycle] 42.010.048 Avg. Issued Instructions Per Scheduler [inst] 29.76

Long Scoreboard

Est. Speedup: 45.97%

On average, each warp of this workload spends 10.9 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the target. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 46.0% of the total average of 23.7 cycles between issuing two instructions.

Key Performance Indicators

Warp Stall

Check the [Warp Stall Sampling \(All Samples\)](#) table for the top stall locations in your source based on sampling data. The [Profiling Guide](#) provides more details on each stall reason.

Warp State (All Cycles)

Stall Long Scoreboard

Stall Barrier

Stall Wait

Stall Math Pipe Throttle

Stall Not Selected

Stall Short Scoreboard

Selected

Stall Dispatch Stall

Stall MIO Throttle

Stall Branch Resolving

Stall IMC Miss

Stall No Instruction

Stall Drain

Stall LG Throttle

Stall Misc

Stall Tex Throttle

Stall Membar

Stall Sleeping

Cycles per Instruction

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that Instructions/Opcode and Executed Instructions are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst] 4,299,776 Avg. Executed Instructions Per Scheduler [inst] 44,789.33

Issued Instructions [inst] 4,210,048 Avg. Issued Instructions Per Scheduler [inst] 44,905.71

FP32 Non-Fused Instructions

This kernel executes 8192 fused and 8192 non-fused FP32 instructions. By converting pairs of non-fused instructions to their [fused](#), higher-throughput equivalent, the achieved FP32 performance could be increased by up to 25%.

Est. Speedup: 4.35%

Key Performance Indicators

Executed Instruction Categories

Integer

Control

Load/Store

Movement

Miscellaneous

Floating Point

Uniform Datapath

Conversion

Executed Warp-Level Instructions/Opcode

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID GPU Name NUMA ID by CPU Affinity CPU Affinity NUMA ID by Memory Affinity

| 0 | NVIDIA GeForce RTX 4060 Laptop GPU | 0 | 0-15 | 0 |

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size 64 Function Cache Configuration CachePreferNone

Registers Per Thread [register/thread] 30 Static Shared Memory Per Block [byte/block] 10,776.14

Block Size 512 Dynamic Shared Memory Per Block [byte/block] 65,536

Threads [thread] 32,768 Driver Shared Memory Per Block [byte/block] 1,02

Waves Per SM 0.89 Shared Memory Configuration Size [Kbyte] 65.54

Uses Green Context 0 Stack Size 1,024

SMs [SM] 24 # TPCs 12

Enabled TPC IDs all -

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Active Cycles [%] 100 Block Limit Registers [block] 4

Theoretical Active Cycles [%] 48 Block Limit Shared Mem [block] 5

Theoretical Active Cycles [%] 87.32 Block Limit Waves [block] 3

Achieved Occupancy [%] 41.92 Block Limit SM [block] 24

Achieved Occupancy [%] 12.04%

The difference between calculated theoretical (100.0%) and measured achieved occupancy (87.3%) can be the result of warp scheduling overheads and workload changes during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the [Launch and Profiling Guide](#) for more details on optimizing occupancy.

Key Performance Indicators

Impact of Varying Register Count Per Thread

% Occupancy

Registers Per Thread

Impact of Varying Block Size

% Occupancy

Block Size

Impact of Varying Shared Memory Usage Per Block

% Occupancy

Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM.

Average SM Active Cycles [cycle] 102,012.88 Average L1 Active Cycles [cycle] 102,012.88

Average L2 Active Cycles [cycle] 64,654.44 Average SMSP Active Cycles [cycle] 102,776.14

Average DRAM Active Cycles [cycle] 49,324 Total SM Elapsed Cycles [cycle] 2,518,000

Total L1 Elapsed Cycles [cycle] 10,072,000 Total L2 Elapsed Cycles [cycle] 1,671,872

Total SMSP Elapsed Cycles [cycle] 2,072,576

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	102,012.88	100,316	103,025	2,448,309
SMSP Active Cycles	102,776.14	101,287	104,908	9,866,509
L1 Active Cycles	102,012.88	100,316	103,025	2,448,309
L2 Active Cycles	64,654.44	63,266	66,048	1,034,471
DRAM Active Cycles	49,324	49,280	49,424	197,296

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst] 363,520 Branch Efficiency [%] 100

Branch Instructions Ratio [%] 0.08 Avg. Divergent Branches [branches] 0

Uncoalesced Global Accesses

This kernel has uncoalesced global accesses resulting in a total of 32768 excessive sectors (10% of the total 327680 sectors). Check the [L2 Theoretical Sectors Global Excessive](#) table for the primary source locations. The [CUDA Occupancy Guide](#) has additional information on reducing uncoalesced data memory accesses.

Est. Speedup: 6.19%

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location

Value

Value (%)

0x7882772722a0 in gemm_wmma_kernel 4,096 19

0x7882772722a0 in gemm_wmma_kernel 4,096 19

0x7882772722a0 in gemm_wmma_kernel 4,096 19

0x7882772722a0 in gemm_wmma_kernel 4,096 19

0x7882772722c0 in gemm_wmma_kernel 4,096 19

Warp Stall Sampling (All Samples)

Location

Value

Value (%)

0x788277272290 in gemm_wmma_ke 495 28

0x788277272790 in gemm_wmma_ke 432 25

0x7882772722a0 in gemm_wmma_ke 195 1

0x788277272200 in gemm_wmma_ke 141 6

0x788277272770 in gemm_wmma_ke 102 8

Most Instructions Executed

Location

Value

Value (%)

0x788277272290 in gemm_wmma_ke 65,536 4

0x788277272290 in gemm_wmma_ke 65,536 4

0x788277272290 in gemm_wmma_ke 65,536 4

0x788277272290 in gemm_wmma_ke 65,536 4

0x788277272290 in gemm_wmma_ke 65,536 4

Follow the [rules outputs](#) to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel. You could also disable [individual sections](#) to focus on selected performance aspects and make profiling faster.